SI 315

Models of Social Information Theory

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1 Course Logistics

1.1 Participation and Attendance

- Attendance is required at lectures and discussions.
- iClicker credit depends on both participation and correctness.

1.2 Assignments and Exams

- Assignment submissions must not be AI-generated.
- There are two exams; the final exam is non-cumulative.
- The lowest two grades are dropped in each category: assignments, quizzes, discussion attendance, and lecture attendance.

2 Introduction to Networks

2.1 Strong and Weak Ties

- Strong ties: frequent interaction, trust, emotional closeness
- Weak ties: infrequent interaction, limited intimacy
- Strong ties: overlapping social circles
- Weak ties: bridges between disconnected groups
- Weak ties: novel information and opportunities

2.2 Bridges and Local Bridges

- Bridge: an edge whose removal disconnects the network into separate components
- Local bridge: an edge whose endpoints do not have any direct ties to the same neighboring nodes
- Local bridge: removing it increases the distance between its endpoints, giving it a span of at least 3
- Span of a bridge: the distance between the endpoints after the edge is removed
- Bridges and local bridges: often connect otherwise distant or disconnected regions of the network

2.3 Clustering

- Clustering: connected neighbors also connected to one another
- Local clustering coefficient: connected neighbor pairs divided by possible neighbor pairs
- Formula: $C_i = \frac{2e_i}{k_i(k_i-1)}$
- e_i: edges among node i's neighbors
- Network clustering coefficient: average local coefficient

2.4 Strong Triadic Closure Property

- Two strong ties sharing a node: likely third edge
- Strong ties around a node: mutually connected neighbors
- Open triad: two strong ties without a connecting edge
- Open triads: pressure toward closure
- Closed triads: higher clustering coefficients

2.5 Homophily

- Homophily: the tendency for people to form ties with similar others
- Similarity may involve age, gender, education, location, interests, or beliefs
- Homophily can produce clustered groups with shared characteristics
- Network ties can reinforce existing similarities through social influence
- Observed similarity may result from selection, influence, or both
- Let p and q be the proportions of two groups, with $p + q = 1$
- Under random mixing, the probability of a cross-group tie is $2pq$
- Cross-group tie probability c under random mixing: expected to be $2pq$
- Homophily where the observed cross-group tie probability is less than expected: $c < 2pq$

2.6 Affiliation Networks

- Affiliation network: a network connecting people to the groups, events, organizations, or activities they participate in
- Two-mode network: contains two types of nodes, such as individuals and affiliations
- One-mode projection: connects two people when they share an affiliation
- Shared affiliations can create indirect social ties and opportunities for interaction
- Affiliations can reveal overlapping communities and patterns of participation

2.7 Social Affiliation Networks

- Social affiliation network: represents relationships between people and the groups, events, or organizations with which they are affiliated
- Membership links connect each person to the affiliations they share
- Shared affiliations can create indirect opportunities for social interaction and information flow
- One-mode projection: connects people who share at least one affiliation
- Projection ties may be weighted by the number of affiliations two people share
- Affiliation networks can reveal overlapping communities, central participants, and patterns of group involvement

2.7.1 Closure in Social Affiliation Networks

- Triadic: nodes with a common neighbor tend to form a direct connection, closing the triad
- Group/focal: nodes connected to the same group or affiliation tend to form direct connections, closing the group-based triad
- Membership: nodes are more likely to join groups that their neighbors are already members of

2.8 Signed Networks

- Signed network: a network in which edges have positive or negative signs, representing friendly or antagonistic relationships
- Positive tie: indicates friendship, trust, or alliance
- Negative tie: indicates enmity, distrust, or conflict
- Balance theory: suggests that triads tend to evolve toward a balanced state, where the product of the signs of the edges is positive
- Balanced triads: all positive edges or two negative edges and one positive edge
- Unbalanced triads: one negative edge and two positive edges or all negative edges

3 Balance

3.1 Balance Theory

- Balance theory: suggests that triads tend to evolve toward a balanced state, where the product of the signs of the edges is positive
- Balanced triads: 3 positive edges or 1 positive edge
- Unbalanced triads: one negative edge and two positive edges or all negative edges
- Balanced network: all triads are balanced

3.2 Weak Balance Theorem

- Weak balance theorem: in a signed network, if all triads are weakly balanced (no triad has exactly two negative edges), the network can be partitioned into two mutually antagonistic groups with positive ties within each group and negative ties between groups
- Weakly balanced triads: triads that have 3 positive edges, 1 positive edge, or 0 positive edges
- Weakly balanced network: all triads in the network are weakly balanced

4 Game Theory

4.1 Strategic-Form Games

- Game: players, available strategies, and payoffs for each strategy profile
- Strategy profile: one strategy chosen by each player
- Payoff: value or outcome a player receives from a strategy profile
- Best response: strategy that maximizes a player's payoff given the other players' strategies
- Strict best response: strategy that gives a strictly higher payoff than every alternative, given the other players' strategies
- More than one best response possible when strategies tie for the highest payoff

4.2 Dominant Strategies

- Dominant strategy: best response to every possible strategy profile of the other players
- Strictly dominant strategy: gives a strictly higher payoff than every alternative for every possible strategy profile of the other players
- Strictly dominant strategy: rational choice regardless of what the other players do
- A game may have no dominant strategy, or may have dominant strategies for some or all players

4.3 Example Games

4.3.1 Presentation vs. Exam

- Compare each player's available choices and payoffs in every outcome
- Mark each player's best response to each choice of the other player
- Use the payoff matrix to determine whether choices align or conflict

4.3.2 Prisoner's Dilemma

- Two players independently choose to cooperate or defect
- Defecting: strictly dominant strategy for each player
- (Defect, Defect): Nash equilibrium, despite mutual cooperation giving both players a better outcome
- Individual incentives can produce an outcome worse for both players than cooperation

4.3.3 A New Product

- Analyze the available choices and payoffs for each player using the lecture payoff matrix
- Check how one player's best response changes with the other player's choice
- Use best responses to identify any Nash equilibria

4.3.4 Three Clients

- Identify the players, available strategies, and payoffs from the lecture example
- For each player, compare payoffs across their choices while holding the other players' choices fixed
- Mark mutual best responses to locate Nash equilibria

4.4 Nash Equilibrium

- Two-player strategy profile (S, T): Nash equilibrium if S is a best response to T and T is a best response to S
- General case: every player's strategy is a best response to the strategies chosen by all other players
- At equilibrium: no player can improve their payoff by changing strategy alone
- Finite-game existence theorem: every game with finitely many players and finitely many strategies has at least one Nash equilibrium in mixed strategies
- Pure-strategy Nash equilibrium: equilibrium using a single strategy per player; not guaranteed to exist

4.5 Focal Points

- Focal point: salient outcome that helps players coordinate without communication
- Salience may come from conventions, shared expectations, or a distinctive feature of the game
- Focal points can help players select among multiple coordination-game equilibria

4.6 Mixed Strategies

- Mixed strategy: probability distribution over a player's available pure strategies
- Pure strategy: one action chosen with probability 1
- Expected payoff: probability-weighted average payoff across possible outcomes
- For strategy profile (s_i, s_{-i}) drawn from mixed strategies:

$$\mathbb{E}[u_i] = \sum_{s_i, s_{-i}} \Pr(s_i) \Pr(s_{-i}) u_i(s_i, s_{-i})$$

- Mixed-strategy Nash equilibrium: each player's probability distribution is a best response to the others' distributions
- In equilibrium, every pure strategy used with positive probability gives the player the same expected payoff

4.6.1 Finding a Mixed-Strategy Equilibrium

- Assign probabilities to one player's strategies
- Choose probabilities that make the other player indifferent among their pure strategies
- Repeat in the opposite direction to find the first player's equilibrium probabilities
- Indifference makes mixing optimal; a player would otherwise put all probability on a higher-payoff pure strategy

4.6.2 Solving for an Unknown Payoff

- **Indifference principle**: in a mixed-strategy NE, each player's mix makes the *other* player indifferent among strategies played with positive probability
- Steps:
  1. Get equilibrium probabilities (usually given)
  2. Write indifference condition for the player whose payoffs are known: uses the opponent's mix
  3. Write indifference condition for the player with the unknown payoff x: $\mathbb{E}[u_1(U)] = \mathbb{E}[u_1(D)]$
  4. Solve for x: linear, one equation, one unknown
- Unknown payoff of player i appears in i's indifference condition, which uses the *opponent's* probabilities
- If x also affects the opponent's mix: use both indifference equations with given probabilities
- If no probabilities given: treat p, q as unknowns; two indifference equations, solve for p, q, x
- Check: all probabilities in $[0, 1]$; no profitable deviation to a pure strategy outside the mix

Example

- Payoffs (Player 1, Player 2):

	L	R
U	$x, 2$	$0, 1$
D	$1, 0$	$3, 3$

- Given: Player 2 plays L with probability $q = \frac{3}{4}$
- Player 1 indifferent between U and D:

$$\mathbb{E}[u_1(U)] = qx + (1-q)\cdot 0 = \tfrac{3}{4}x$$
$$\mathbb{E}[u_1(D)] = q\cdot 1 + (1-q)\cdot 3 = \tfrac{3}{4} + \tfrac{3}{4} = \tfrac{3}{2}$$

- Set equal: $\frac{3}{4}x = \frac{3}{2}$, so $\boxed{x = 2}$

4.6.3 Probabilities from a Complete Payoff Matrix

- p = Pr(Player 1 plays U); q = Pr(Player 2 plays L)
- p comes from **Player 2's** indifference; q comes from **Player 1's** indifference
- Own payoffs only determine what the opponent does; given q is not needed to find p
- Check dominated strategies first: strictly dominated gets probability 0; fully mixed NE needs no dominant strategy
- Valid only if $0 < p, q < 1$

Example

- Payoffs (Player 1, Player 2):

	L	R
U	$4, 3$	$1, 1$
D	$2, 0$	$5, 4$

- Find p: Player 2 indifferent between L and R

$$\mathbb{E}[u_2(L)] = 3p + 0(1-p) = 3p$$
$$\mathbb{E}[u_2(R)] = 1p + 4(1-p) = 4 - 3p$$

- $3p = 4 - 3p$, so $\boxed{p = \frac{2}{3}}$
- Find q: Player 1 indifferent between U and D

$$\mathbb{E}[u_1(U)] = 4q + 1(1-q) = 1 + 3q$$
$$\mathbb{E}[u_1(D)] = 2q + 5(1-q) = 5 - 3q$$

- $1 + 3q = 5 - 3q$, so $\boxed{q = \frac{2}{3}}$
- Mixed NE: $p = q = \frac{2}{3}$
- Sanity check: $q = \frac{1}{2}$ is not an equilibrium value; $\mathbb{E}[u_1(U)] = 2.5 < 3.5 = \mathbb{E}[u_1(D)]$, so Player 1 strictly prefers D

4.7 Matching Pennies

- Each player chooses Heads or Tails
- Player 1 wins when choices match; Player 2 wins when choices differ
- No pure-strategy Nash equilibrium: either player can switch and improve their payoff
- Let q be the probability Player 2 chooses Heads
- Player 1's expected payoff from Heads: $q - (1-q) = 2q - 1$
- Player 1's expected payoff from Tails: $(1-q) - q = 1 - 2q$
- Player 1 is indifferent when $2q - 1 = 1 - 2q$, so $q = \frac{1}{2}$
- By symmetry, Player 1 also chooses Heads with probability $\frac{1}{2}$
- Mixed-strategy Nash equilibrium: each player chooses Heads and Tails with probability $\frac{1}{2}$

4.8 Coordination Game

- Players benefit from choosing strategies that match or coordinate
- Coordination games often have multiple Nash equilibria
- Equilibrium selection: players need expectations, conventions, or focal points to coordinate on one equilibrium
- A coordinated outcome may be better for everyone than a mismatched outcome

4.9 Network Traffic

- Players choose routes in a network to minimize their own travel time
- Travel time on a route depends on the number of players using it
- Nash equilibrium occurs when no player can reduce their travel time by unilaterally changing routes
- Equilibria may be inefficient compared to a centrally coordinated solution (Braess's paradox)

5 Auctions

- Types of auctions: ascending (English), descending (Dutch), first-price sealed-bid, second-price sealed-bid (Vickrey)
- Second-price sealed-bid auction: buyers are players, bids are strategies—assume each buyer has a true value v_i and submits a bid b_i; highest bidder "payoff" is $v_i - b_j$ (the money saved compared to their true value), where b_j is the second-highest bid; dominant strategy is to bid their true value $b_i = v_i$; truthful bidding leads to efficient allocation of the item.

5.1 Bipartite Matching

- Bipartite graph: graph whose vertices can be divided into two disjoint sets where no two vertices within the same set are adjacent
- Bipartite (perfect) matching: set of edges in a bipartite graph such that no two edges share a common vertex
- Maximum bipartite matching: bipartite matching that contains the largest possible number of edges
- Constricted set: subset of vertices in one partition of a bipartite graph such that the number of neighbors in the other partition is strictly smaller than the number of vertices in the subset

5.1.1 Weighted Edges

- Weighted bipartite graph: bipartite graph where each edge has an associated weight
- Maximum weight bipartite matching (optimal assignment): bipartite matching that maximizes the sum of the weights of the selected edges

5.1.2 Finding Market-Clearing Prices

- Market-clearing prices: prices assigned to edges in a weighted bipartite graph such that each vertex in one partition is matched to a vertex in the other partition with the highest possible weight given the prices, ensuring that the market clears (all supply meets demand); aka, competitive equilibrium prices.
- Preferred-seller graph: bipartite graph constructed based on the current prices, where each vertex in one partition points to its most preferred vertex in the other partition (buyers pick their favorite sellers based on payoffs)
- Iterative procedure to find market-clearing prices:

  1. Initialize prices for all edges, typically starting with zero.
  2. Construct preferred-seller graph based on the current prices, where each vertex in one partition points to its most preferred vertex in the other partition.
  3. If there is no perfect matching, increase the prices of the nodes with multiple preferred edges and repeat the process until a perfect matching is found.

5.2 Restricted Markets

- Power in social networks: the ability of a node to influence or control the flow of information or resources within the network; often related to the node's position and connectivity within the network
- Exchange experiment: experimental setup used to study how individuals trade or exchange resources within a network, often used to analyze the effects of network structure on bargaining power and market outcomes

- Stability in network exchange: a condition where no subset of nodes can rearrange their exchanges to make themselves strictly better off, ensuring that the network exchange structure remains stable over time
  - Unstable where connected nodes not exchanging have a total gain less than the maximum possible gain

6 Network Searching

6.1 Hubs and Authorities

- Used to identify important nodes in a network
- Hubs are nodes that point to many authorities
- Authorities are nodes that are pointed to by many hubs

6.1.1 Measuring Rankings

- Authority score: sum of scores of hubs pointing to the authority
- Hub score: sum of scores of authorities pointed to by the hub
- HITS: iteratively update authority and hub scores
- Instabilities: an unused edge between X and Y when the sum of X's and Y's scores is less than the edge value
- Stable: contains no instabilities
- Nash bargaining solution: if A and B make a deal with best outside options X and Y, respectively, then A receives $X + \frac{S}{2}$ and B receives $Y + \frac{S}{2}$, where $S = V - (X + Y)$ is the surplus from the deal; cannot achieve if $S < 0$
- Balanced: all deals are balanced
- A balanced outcome is also stable

6.1.2 Example Using Adjacency Matrix

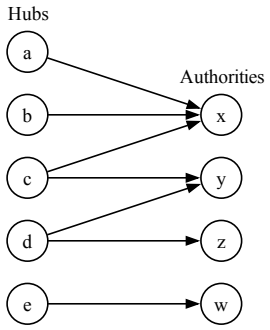


Figure 6.1: Bipartite network of hubs pointing to authorities.

- Node order: hubs (a, b, c, d, e); authorities (x, y, z, w)
- Adjacency matrix A: rows = hubs; columns = authorities

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Initialize hub scores: $\mathbf{h}^{(0)} = (1, 1, 1, 1, 1)$
- Each iteration: $\mathbf{a}^{(t)} = A^{\mathsf{T}}\mathbf{h}^{(t-1)}$; $\mathbf{h}^{(t)} = A\mathbf{a}^{(t)}$
- Integer updates below: no intermediate normalization; reported scores normalized to unit L_2 length

- Iteration 1:

$$\mathbf{a}^{(1)} = A^{\mathsf{T}}\mathbf{h}^{(0)} = (3, 2, 1, 1), \quad \widehat{\mathbf{a}}^{(1)} \approx (0.775, 0.516, 0.258, 0.258)$$
$$\mathbf{h}^{(1)} = A\mathbf{a}^{(1)} = (3, 3, 5, 3, 1), \quad \widehat{\mathbf{h}}^{(1)} \approx (0.412, 0.412, 0.687, 0.412, 0.137)$$

- Iteration 2:

$$\mathbf{a}^{(2)} = A^{\mathsf{T}}\mathbf{h}^{(1)} = (11, 8, 3, 1), \quad \widehat{\mathbf{a}}^{(2)} \approx (0.788, 0.573, 0.215, 0.072)$$
$$\mathbf{h}^{(2)} = A\mathbf{a}^{(2)} = (11, 11, 19, 11, 1), \quad \widehat{\mathbf{h}}^{(2)} \approx (0.409, 0.409, 0.706, 0.409, 0.037)$$

- Iteration 3:

$$\mathbf{a}^{(3)} = A^{\mathsf{T}}\mathbf{h}^{(2)} = (41, 30, 11, 1), \quad \widehat{\mathbf{a}}^{(3)} \approx (0.789, 0.577, 0.212, 0.019)$$
$$\mathbf{h}^{(3)} = A\mathbf{a}^{(3)} = (41, 41, 71, 41, 1), \quad \widehat{\mathbf{h}}^{(3)} \approx (0.408, 0.408, 0.707, 0.408, 0.010)$$

- Iteration 3 ranking: c strongest hub; x strongest authority
- Ranking trend: c and x dominant; e and w increasingly weak

6.2 PageRank

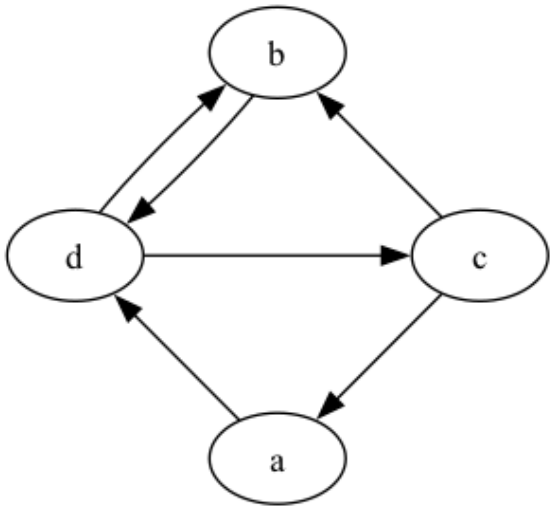
- Method for ranking nodes in a network based on the structure of incoming links
- Counts the number and quality of links to a page to determine its importance
  1. Assign equal scores to all nodes: $\frac{1}{n}$
  2. Update scores based on the scores of incoming links, typically using the formula: $\mathrm{PR}(i) = \sum_{j \in \mathrm{I}(i)} \frac{\mathrm{PR}(j)}{|\mathrm{O}(j)|}$ where $\mathrm{I}(i)$ is the set of nodes linking to i and $|\mathrm{O}(j)|$ is the number of outbound links from node j
- Rankings for iteration i are calculated using rankings from iteration $i-1$, so do not use rankings calculated in the current iteration

6.2.1 Scaled PageRank

- Problem with basic PageRank: a closed cycle can trap all rank; pages outside the cycle receive none
- Fix: model a random surfer who follows a link with probability d, or jumps to a random page with probability $1-d$
- d: damping factor, often 0.85; N: number of pages
- Update each page's rank with link-following score plus a share of the jump probability:

$$\mathrm{PR}_{t+1}(i) = d \sum_{j \in \mathrm{I}(i)} \frac{\mathrm{PR}_t(j)}{|\mathrm{O}(j)|} + \frac{1-d}{N}$$

- Random jumps let rank escape cycles and reach every page; rankings remain well-defined



(a) Network for PageRank.

$$\mathrm{PR}_{t+1}(A) = \frac{\mathrm{PR}_t(C)}{2}$$
$$\mathrm{PR}_{t+1}(B) = \frac{\mathrm{PR}_t(C)}{2} + \frac{\mathrm{PR}_t(D)}{2}$$
$$\mathrm{PR}_{t+1}(C) = \frac{\mathrm{PR}_t(D)}{2}$$
$$\mathrm{PR}_{t+1}(D) = \mathrm{PR}_t(A) + \mathrm{PR}_t(B)$$

(b) Update equations for the PageRank of each node.

Iteration	A	B	C	D
0	0.25	0.25	0.25	0.25
1	0.125	0.25	0.125	0.5
2	0.0625	0.3125	0.25	0.375
3	0.125	0.3125	0.1875	0.375

(c) First 3 iterations of PageRank for the given network.

Figure 6.2: Worked PageRank example.

6.2.2 Other Centrality Measures

- n: number of nodes; $\mathrm{dist}(u, v)$: shortest-path distance from u to v
- **Degree centrality**: number of edges incident to a node; high score means many direct connections

$$C_D(v) = \frac{\deg(v)}{n-1}$$

  for an undirected network; division by $n-1$ gives the normalized score
- **In-degree centrality**: number of edges entering v; $\deg^-(v)$, often a measure of incoming attention or popularity
- **Out-degree centrality**: number of edges leaving v; $\deg^+(v)$, often a measure of outgoing activity

$$C_{\text{in}}(v) = \frac{\deg^-(v)}{n-1}, \qquad C_{\text{out}}(v) = \frac{\deg^+(v)}{n-1}$$

  for a directed network
- **Closeness centrality**: inverse of the average shortest-path distance from v to other nodes; high score means quick reach to others

$$C_C(v) = \frac{n-1}{\sum_{u \neq v} \mathrm{dist}(v, u)}$$

  for a connected network; disconnected networks require a reachable-node variant, with conventions varying

- **Betweenness centrality**: share of shortest paths between node pairs that pass through v; high score means strong brokerage or control of paths

$$C_B(v) = \sum_{\substack{s \neq v,\ t \neq v \\ s \neq t}} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

  where σ_{st}: number of shortest paths from s to t, and $\sigma_{st}(v)$: number passing through v; normalized forms depend on directed/undirected and ordered/unordered pair conventions

7 Advertising and Web Search

- Search ads are unique in that they serve targeted ads with specific interests at the *right time*
- Determine prices for ads with *auctions*
- Value of an ad slot: click-through rate, revenue per click, etc.
- We could make market clearing prices if we know advertisers' valuations, but we generally don't
- 1st-price auction causes underreporting of value, and 2nd-price auction is only tested on single-item auctions
- Incorporate idea of "harm": does a bidder cause harm to other bidders by winning a certain slot?
- Need to compare the total valuation of other bidders to the total valuation if the bidder were to leave the auction
- Difference in total valuations is the value gain if the bidder were to leave the auction
- The winner of the least-valuable slot does not make any effect
- Differences in total valuations are the Vickrey-Clarke-Groves (VCG) prices

7.1 VCG Auctions

7.1.1 Process

1. Buyers submit bids for each item
2. Choose socially-optimal matching (e.g. with market-clearing prices)
3. Charge each buyer the VCG price for their winning item

7.1.2 Properties

- Bidding your true value is the dominant strategy
- Maximizes total valuation of the perfect matching from slots to advertisers

7.2 Real-World Pricing

- Generalized second-price auction
- Buyers submit bids
- Sell each slot i to the ith highest bidder, at the $i+1$st highest bid
- But, bidding true value is *not* dominant strategy